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Linguistic Surveillance in Oncology: A Neural Oscillation Signature Theory Framework for Cancer-Related Cognitive Impairment Detection, Treatment Neurotoxicity Monitoring, and Survivorship Cognitive Tracking

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The author is the founder and president of Beverly Index LLC, a company developing commercial applications based on the theoretical framework presented in this paper. This potential conflict of interest is disclosed in accordance with APA ethical standards. The theoretical framework and proposed study design presented herein are offered for scientific scrutiny and collaborative development by the research community.

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Abstract

Cancer-related cognitive impairment (CRCI) affects 30–75% of patients receiving systemic cancer therapy, yet no scalable, objective, real-time monitoring method currently exists to track cognitive trajectory across the cancer care continuum. This paper proposes a linguistic surveillance framework based on Neural Oscillation Signature Theory (NOST; Beverly, 2026b)

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for continuous cognitive monitoring from pre-treatment baseline through active treatment, post-treatment recovery, and long-term survivorship. Six pathways through which cancer and its treatment affect neural oscillatory activity are identified: (1) cancer-driven systemic inflammation producing pre-treatment cognitive impairment, (2) chemotherapy-induced neurotoxicity disrupting oscillatory dynamics across treatment cycles, (3) radiation-related white matter injury producing progressive oscillatory degradation, (4) immunotherapy-related neurological adverse events generating acute oscillatory perturbation, (5) brain metastases and primary CNS tumors producing focal oscillatory disruption, and (6) endocrine therapy effects on oscillatory function in hormone-sensitive cancers. Because NOST proposes that neural oscillatory states leave detectable signatures in linguistic output, these cancer-related oscillatory changes should manifest as measurable shifts in the statistical properties of language. This paper articulates the detection chain for each pathway, specifies the predicted linguistic signatures, distinguishes cancer-related cognitive impairment from the vascular and cardiac signatures described in companion papers, and proposes a prospective longitudinal validation study across the cancer care continuum. If confirmed, the framework would establish language-based monitoring as the first objective, scalable, real-time cognitive surveillance tool in oncology—addressing a critical unmet need affecting millions of cancer patients and survivors.
Keywords: cancer-related cognitive impairment, chemobrain, neurotoxicity monitoring, linguistic markers, NOST, NOSS, oncology, survivorship, chemotherapy, immunotherapy, cognitive surveillance

Linguistic Surveillance in Oncology:

A Framework for Cancer-Related Cognitive Monitoring

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Cancer-related cognitive impairment (CRCI)—colloquially known as “chemobrain”—is among the most prevalent, most distressing, and least well-managed consequences of cancer and its treatment. Between 30% and 75% of patients receiving systemic chemotherapy experience objectively measurable cognitive decline during treatment (Janelsins et al., 2014), and longitudinal studies document persistent cognitive deficits in 20–35% of survivors years after treatment completion (Koppelmans et al., 2012). The affected domains—processing speed, executive function, attention, and episodic memory—are precisely those most central to occupational function, daily independence, and quality of life.

Despite its prevalence and impact, CRCI remains poorly monitored in clinical practice. Standard oncological care includes extensive monitoring of hematological toxicity (complete blood counts), hepatic toxicity (liver function tests), cardiac toxicity (ejection fraction), and nephrotoxicity (creatinine clearance)—but cognitive toxicity receives no routine objective assessment. When cognitive concerns are addressed, it is typically through patient self-report or sporadic neuropsychological testing administered at multi-month intervals, if at all. No continuous, objective, scalable cognitive monitoring system currently exists in oncology.

This paper proposes that Neural Oscillation Signature Theory (NOST; Beverly, 2026b) provides the theoretical foundation for such a system. NOST establishes that neural oscillatory states systematically shape language production, creating detectable statistical signatures in spoken and written text. Because cancer and its treatment affect neural oscillatory activity through multiple well-documented pathways, those effects should produce measurable linguistic changes amenable to continuous computational monitoring. The result would be the first real-time cognitive surveillance tool in oncology—one capable of tracking cognitive trajectory from pre-treatment baseline through active treatment, recovery, and long-term survivorship.

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Companion papers have applied the NOST framework to subclinical cerebrovascular detection (Beverly, 2026e) and cardiovascular pathology detection through cardiocerebral coupling (Beverly, 2026f). The present paper addresses a distinct clinical domain—oncology—with distinct pathophysiology, distinct clinical populations, and distinct intervention pathways. While the final links of the detection chain (oscillatory disruption → linguistic signature change → computational detection) are shared across all applications, the entry points, temporal dynamics, and clinical decision logic differ substantially.

Six Pathways: Cancer-Related Neural Oscillatory Disruption

Cancer and its treatment affect cerebral neural oscillatory activity through six identifiable pathways, each with distinct mechanisms, temporal dynamics, and clinical implications.

Pathway 1: Cancer-Driven Systemic Inflammation

The first pathway operates before any treatment is administered. Tumors establish a systemic inflammatory microenvironment through the production of pro-inflammatory cytokines—interleukin-6 (IL-6), tumor necrosis factor-alpha (TNF-α), interleukin-1β (IL-1β), and C-reactive protein (CRP)—that circulate systemically and cross the blood-brain barrier through multiple mechanisms including active transport, circumventricular organ access, and inflammation-induced barrier permeability (Dantzer et al., 2008). The resulting neuroinflammation produces measurable cognitive deficits in a substantial proportion of patients prior to treatment initiation.

Wefel et al. (2011) documented that approximately 30% of breast cancer patients exhibited objectively measurable cognitive impairment at baseline, before any systemic therapy. Similar findings have been reported in lung, colorectal, and hematological malignancies (Lange et al., 2019). The cognitive profile of cancer-related pre-treatment impairment is consistent with

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inflammation-mediated neuronal dysfunction: reduced processing speed, diminished executive function, and impaired attention—with relative preservation of crystallized abilities and overlearned skills.

The electroencephalographic correlates of cytokine-mediated cognitive impairment include: increased theta-band power reflecting hippocampal and medial temporal dysfunction, reduced alpha-band reactivity reflecting compromised attentional gating, and slowed dominant frequency reflecting global cortical compromise (Lasselin et al., 2020; Harrison et al., 2014). These are the same oscillatory changes that NOST is designed to detect through linguistic analysis. The predicted NOSS activation pattern for cancer-driven pre-treatment cognitive impairment is: primary S4 (Neurocognitive) elevation with processing-speed-predominant features, with possible secondary S1 (Mood) elevation reflecting the well-documented association between pro-inflammatory cytokines and depressive symptomatology (Miller & Raison, 2016).

The clinical value of detecting pre-treatment cognitive impairment is substantial: it establishes a true cognitive baseline that accounts for cancer-related effects already present at diagnosis, identifies patients at elevated risk for treatment-related cognitive decline (pre-treatment impairment is the single strongest predictor of post-treatment impairment; Ahles et al., 2010), and enables pre-treatment neuroprotective intervention in high-risk patients.

Pathway 2: Chemotherapy-Induced Neurotoxicity

Systemic chemotherapy produces cognitive impairment through multiple neurotoxic mechanisms: direct neuronal damage from agents that cross the blood-brain barrier (methotrexate, cytarabine, ifosfamide, 5-fluorouracil), oxidative stress-mediated neuronal injury, chemotherapy-amplified neuroinflammation through microglial activation and additional

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cytokine release, disruption of hippocampal neurogenesis, white matter demyelination, and telomere shortening with accelerated cellular aging in neural tissue (Ahles & Root, 2018; Dietrich et al., 2006).

The cognitive consequences are dose-dependent and temporally structured. Most patients experience measurable cognitive decline during active treatment, with nadir typically occurring at the midpoint of multi-cycle regimens. The characteristic pattern is acute cognitive perturbation following each treatment cycle, with partial recovery between cycles and a cumulative trajectory of progressive cognitive decline over the course of treatment (Jim et al., 2012). This cyclic pattern is a critical feature for NOST-based monitoring: it produces a temporally predictable oscillatory signature—periodic perturbation synchronized with treatment cycles superimposed on a longer-term trajectory—that is pathognomonic for chemotherapy-related neurotoxicity and distinguishable from all other causes of cognitive decline.

The documented oscillatory changes during chemotherapy include: increased delta and theta power during the acute post-treatment period, reduced alpha power and peak frequency, diminished event-related synchronization during cognitive tasks, and disrupted cross-frequency coupling between frontal theta and posterior alpha bands (Ahles & Root, 2018; de Ruiter et al., 2011). The degree of oscillatory disruption correlates with the severity of cognitive impairment and with inflammatory biomarker levels.

NOST predicts that chemotherapy-induced neurotoxicity should produce a distinctive temporal-linguistic signature: acute post-cycle perturbation (reduced syntactic complexity, increased processing pauses, diminished information density, impaired semantic coherence) followed by partial between-cycle recovery, with progressive baseline degradation across the treatment course. The magnitude of each perturbation and the degree of inter-cycle recovery

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provide real-time information about cumulative neurotoxicity—information that could inform treatment modification decisions.

Pathway 3: Radiation-Related White Matter Injury

Cranial radiation therapy produces progressive white matter injury through demyelination, axonal loss, and microvasculopathy in the irradiated volume and, to a lesser extent, in the surrounding normal brain tissue (Greene-Schloesser et al., 2012). Unlike chemotherapy-induced cognitive impairment, which typically manifests during or shortly after treatment, radiation-related cognitive decline often follows a delayed trajectory: relatively preserved function during and immediately after radiation, followed by progressive cognitive deterioration emerging months to years later as white matter damage accumulates.

The cognitive profile of radiation-related injury is distinguished by its anatomical specificity: whole-brain radiation produces diffuse cognitive decline affecting multiple domains, while focal radiation (stereotactic radiosurgery, partial brain fields) produces deficits corresponding to the irradiated territory. Temporal lobe radiation preferentially impairs memory. Frontal radiation preferentially impairs executive function. Hippocampal-avoidance radiation techniques were developed precisely to preserve the cognitive functions most vulnerable to radiation injury (Gondi et al., 2014).

The oscillatory consequences of radiation-related white matter injury include: progressive increase in slow-wave (delta and theta) power in the irradiated region, reduced inter-regional coherence reflecting disrupted white matter connectivity, and slowed processing speed reflected in delayed event-related potential components. These changes are progressive and dose-dependent, making them particularly suitable for longitudinal trajectory monitoring.

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NOST predicts that radiation-related cognitive decline should produce a slowly progressive linguistic degradation trajectory—distinct from the cyclic pattern of chemotherapy neurotoxicity and the acute onset of brain metastasis. The linguistic features affected would be determined by the radiation field: temporal radiation producing memory-dependent linguistic changes (reduced autobiographical specificity, impaired temporal reference accuracy), frontal radiation producing executive-linguistic changes (reduced syntactic complexity, impaired discourse planning), and whole-brain radiation producing diffuse multi-domain linguistic decline. This anatomical specificity of the predicted linguistic signature, if confirmed, would provide treatment-relevant information about the location and progression of radiation-related injury.

Pathway 4: Immunotherapy-Related Neurological Adverse Events

Immune checkpoint inhibitors (anti-PD-1, anti-PD-L1, anti-CTLA-4) have transformed cancer treatment but produce a spectrum of immune-related adverse events (irAEs), including neurological complications affecting 1–5% of patients. Neurological irAEs range from mild (headache, peripheral neuropathy) to severe (autoimmune encephalitis, aseptic meningitis, transverse myelitis, myasthenia gravis-like syndrome), and they can be life-threatening if not detected early (Cuzzubbo et al., 2017).

The detection challenge is significant: neurological irAEs can develop insidiously, with initial symptoms—subtle cognitive slowing, mild confusion, personality change—easily attributed to cancer-related fatigue, depression, or the general stress of cancer treatment. By the time overt neurological symptoms trigger clinical evaluation, substantial immune-mediated neural damage may have occurred. Early detection is critical because immunotherapy-related neurological adverse events are often reversible with prompt immunosuppressive treatment

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(corticosteroids, IVIG, plasmapheresis), but delayed treatment leads to permanent neurological deficit (Haanen et al., 2017).

The predicted oscillatory signature of immunotherapy-related encephalitis differs from chemotherapy neurotoxicity in its temporal profile: rather than the cyclic perturbation synchronized with treatment cycles, immunotherapy-related neurological adverse events typically produce a subacute, progressive deterioration that develops over days to weeks, often beginning 6–12 weeks after treatment initiation. The oscillatory changes in autoimmune encephalitis include diffuse or focal slowing, epileptiform activity, and disrupted normal background rhythms (Graus et al., 2016). NOST-based monitoring during immunotherapy could detect the linguistic manifestations of early-stage neurological irAEs—progressive decline in cognitive-linguistic function not synchronized with treatment cycles and not following the expected chemotherapy neurotoxicity pattern—triggering urgent neurological evaluation.

Pathway 5: Brain Metastases and Primary CNS Tumors

Brain metastases occur in 20–40% of patients with systemic cancer, with highest rates in lung cancer, breast cancer, and melanoma (Nayak et al., 2012). Primary CNS tumors—gliomas, meningiomas, CNS lymphoma—directly disrupt local neural circuit function. Both produce focal oscillatory disturbance in tumor-adjacent cortex: increased delta and theta power, reduced beta and gamma activity, and disrupted functional connectivity between the affected region and its network partners (Derks et al., 2018).

The clinical detection challenge for brain metastases is that many are initially asymptomatic or produce symptoms too subtle for patients to report. Current surveillance protocols rely on periodic contrast-enhanced MRI at multi-month intervals. Between scans, metastatic deposits may grow, produce vasogenic edema, and cause progressive neural damage

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before detection. NOST-based linguistic monitoring could supplement imaging surveillance by detecting the functional consequences of developing brain metastases between scheduled scans. A focal linguistic change—for example, an acute decline in naming accuracy suggesting temporal lobe involvement, or a selective executive function decline suggesting frontal involvement—could trigger early imaging and treatment.

The distinguishing feature of the brain metastasis linguistic signature, relative to other oncological pathways, is its focal rather than diffuse character, its acute or subacute onset (distinct from the gradual trajectory of radiation injury or the cyclic pattern of chemotherapy toxicity), and its potential lateralization if the metastasis affects language-dominant hemisphere structures. In high-risk patient populations, NOST monitoring specifically tuned for focal cognitive change could provide a functional surveillance layer complementing structural MRI surveillance.

Pathway 6: Endocrine Therapy Effects

Endocrine therapies used in hormone-sensitive cancers—aromatase inhibitors (AIs) and selective estrogen receptor modulators (SERMs) in breast cancer, androgen deprivation therapy (ADT) in prostate cancer—produce cognitive effects through hormonal mechanisms distinct from the neurotoxic and inflammatory pathways of cytotoxic chemotherapy.

Estrogen is a neuroactive hormone with well-documented effects on neural oscillatory function: it modulates hippocampal theta rhythm, enhances alpha-band reactivity, and supports cholinergic neurotransmission underlying attentional and memory processes (Grigorova et al., 2006). Estrogen deprivation through aromatase inhibition produces measurable cognitive effects—particularly in verbal memory and processing speed—in a substantial proportion of breast cancer patients (Bender et al., 2007). Similarly, testosterone has neuroactive properties,

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and androgen deprivation in prostate cancer produces documented cognitive decline in spatial ability, verbal memory, and executive function (Gonzalez et al., 2015).

The oscillatory consequences of endocrine deprivation differ from those of cytotoxic neurotoxicity: estrogen deprivation preferentially affects alpha and theta dynamics in temporal and hippocampal circuits, while testosterone deprivation affects spatial processing networks and fronto-parietal executive circuits. NOST predicts that endocrine therapy-related cognitive decline should produce a distinctive NOSS pattern: memory-predominant features in estrogen-deprived patients (reduced autobiographical specificity, impaired temporal reference, vocabulary access difficulty) and spatial-executive features in androgen-deprived patients (reduced syntactic complexity, impaired sequential planning). These patterns differ from the processing-speed-predominant pattern of chemotherapy neurotoxicity and the focal pattern of brain metastasis, enabling differential attribution within the oncological context.

The clinical relevance is direct: endocrine therapies are administered for years (typically 5–10 years in breast cancer), and the cognitive effects are a primary reason patients discontinue treatment despite survival benefits. Objective monitoring of endocrine therapy-related cognitive trajectory could inform shared decision-making about treatment continuation, dose modification, or switch to alternative endocrine agents.

NOST Across the Cancer Care Continuum

A distinguishing feature of the present framework, relative to the cerebrovascular and cardiac applications described in companion papers, is its applicability across the entire cancer care trajectory. The six pathways described above activate at different phases of the cancer experience, creating a temporally structured surveillance framework.

Phase 1: Pre-Treatment Baseline

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At the point of cancer diagnosis, prior to treatment initiation, NOST-based assessment establishes the patient's cognitive-linguistic baseline. This baseline serves multiple functions: it detects pre-existing cancer-related cognitive impairment (Pathway 1), identifies patients with elevated vulnerability to treatment-related cognitive decline, provides the individualized reference against which all subsequent trajectory changes are measured, and captures baseline prosodic and autonomic features that enable differentiation of cancer-related from cardiac-origin cognitive changes (as described in Beverly, 2026f). Baseline assessment requires a minimum of three standardized language samples collected over a two-week period to establish reliable individual norms.

Phase 2: Active Treatment Monitoring

During active treatment—chemotherapy, radiation, immunotherapy, endocrine therapy, or combinations thereof—NOST monitoring shifts to high-frequency mode. Weekly or bi-weekly language samples enable detection of: acute chemotherapy neurotoxicity correlated with treatment cycles (Pathway 2), progressive radiation-related cognitive decline emerging during or shortly after cranial radiation (Pathway 3), immunotherapy-related neurological adverse events developing over weeks (Pathway 4), endocrine therapy effects accumulating over months (Pathway 6), and brain metastasis development between scheduled imaging surveillance (Pathway 5).

The treatment-cycle-aware monitoring capability is critical. The system must distinguish expected treatment-related cognitive perturbation (which, within limits, is an acceptable consequence of life-saving therapy) from unexpected or excessive perturbation that signals neurotoxicity requiring clinical intervention. This requires modeling the expected cognitive trajectory for each treatment regimen—the predicted timing, magnitude, and duration of

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treatment-cycle-related cognitive perturbation—and flagging patients whose actual trajectory exceeds the expected bounds.

Phase 3: Post-Treatment Recovery

Following treatment completion, NOST monitoring tracks the cognitive recovery trajectory. Most patients show measurable cognitive improvement in the 6–12 months following treatment completion, but the rate and completeness of recovery varies substantially. Patients whose recovery trajectory is slower than expected, plateaus prematurely, or reverses after initial improvement represent distinct clinical populations: slow recovery may reflect persistent neuroinflammation amenable to anti-inflammatory intervention, plateau may reflect established white matter damage requiring cognitive rehabilitation, and reversal may signal disease recurrence, late-onset radiation effect, or development of a secondary neurodegenerative process.

Phase 4: Long-Term Survivorship

The survivorship monitoring application addresses one of the most significant unmet needs in cancer care. There are currently over 18 million cancer survivors in the United States alone (American Cancer Society, 2022), and cognitive complaints rank among their most frequent and most functionally disabling long-term concerns. Longitudinal studies document persistent cognitive deficits in 20–35% of survivors five or more years after treatment completion (Koppelmans et al., 2012).

Three specific survivorship surveillance applications are proposed. First, persistent CRCI monitoring: tracking cognitive trajectory in survivors with ongoing cognitive complaints to distinguish stable impairment (which may benefit from cognitive rehabilitation) from progressive impairment (which requires medical investigation for recurrence, secondary neurodegeneration, or treatable comorbidities). Second, late-onset cognitive effects: detecting cognitive decline

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emerging years after treatment completion—a documented phenomenon in survivors of childhood cancer and recipients of cranial radiation (Armstrong et al., 2016). Third, accelerated aging surveillance: cancer treatment, particularly alkylating chemotherapy and radiation, is associated with accelerated biological aging including shortened telomeres, epigenetic aging, and elevated inflammatory markers (Henderson et al., 2014). NOST monitoring could detect the cognitive correlates of accelerated aging—a trajectory of cognitive decline that proceeds faster than the expected age-related trajectory—triggering evaluation for modifiable contributors.

Differential Signature: Cancer-Related Versus Other Etiologies

The clinical utility of NOST-based oncological surveillance depends critically on the ability to distinguish cancer-related cognitive changes from other causes of cognitive decline that may co-occur in cancer patients, including: age-related cognitive change (most cancer patients are older adults), cerebrovascular disease (sharing risk factors with cancer), cardiac disease (cardiotoxicity is a recognized consequence of several cancer treatments), depression and anxiety (highly prevalent in cancer populations), and primary neurodegenerative disease (coincidental but requiring different management).

The present framework proposes that cancer-related cognitive changes are distinguishable from these alternative etiologies through three discriminating features.

First, temporal synchrony with cancer treatment. Chemotherapy-induced cognitive decline is synchronized with treatment cycles. No other cause of cognitive decline produces this cyclic temporal pattern. A linguistic trajectory showing periodic perturbation aligned with chemotherapy administration is essentially diagnostic of treatment-related neurotoxicity, regardless of overlap with other etiological signatures.

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Second, absence of prosodic-autonomic features. The cardiac detection framework (Beverly, 2026f) identifies prosodic changes—reduced pitch variability, altered speech rhythm—as the earliest marker of cardiac autonomic dysregulation. Cancer-related cognitive impairment, except when accompanied by specific cardiotoxic treatment effects, is not predicted to produce the prosodic-autonomic signature because its primary mechanisms (inflammation, direct neurotoxicity, white matter injury) do not target the autonomic innervation of laryngeal and respiratory musculature. The presence or absence of prosodic features therefore provides a discriminating criterion between cardiac-origin and cancer-origin cognitive decline.

Third, domain-specific profiles. Cancer-related cognitive impairment preferentially affects processing speed and attention in chemotherapy-related cases, memory in endocrine-therapy-related cases, and domain-specific function in radiation-field-specific cases. Cerebrovascular disease preferentially affects executive function and processing speed. Alzheimer's-type neurodegeneration preferentially affects episodic memory with early loss of temporal specificity. Depression produces motivational and affective linguistic changes preceding cognitive changes. These domain-specificity differences, while overlapping, provide probabilistic discrimination when combined with temporal and prosodic features.

The Differential Signature Engine described in the NOUS Clinical Surveillance Platform architecture (Beverly, 2026g) would be extended with an oncological template incorporating treatment-cycle temporal analysis, prosodic exclusion criteria, and domain-specific pattern matching. The system would receive treatment schedule data from the EHR (chemotherapy cycle dates, radiation treatment dates, immunotherapy administration dates) to enable treatment-cycle-aware trajectory analysis.

Proposed Validation Study

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Study Design

The proposed study is a prospective longitudinal cohort design following patients from pre-treatment baseline through 24 months post-treatment completion, providing comprehensive coverage of the cancer care continuum. The design is specifically structured to test whether NOST-based linguistic monitoring detects cognitive changes across all six pathways and across all four phases of the cancer care continuum.

Participants

Four cohorts would be recruited: (1) a chemotherapy cohort (n = 200) comprising adults aged 30–75 newly diagnosed with solid tumors (breast, lung, colorectal) scheduled to receive multi-cycle systemic chemotherapy, enrolled prior to treatment initiation; (2) an immunotherapy cohort (n = 100) comprising patients with melanoma, non-small-cell lung cancer, or renal cell carcinoma scheduled to receive immune checkpoint inhibitor therapy; (3) an endocrine therapy cohort (n = 100) comprising breast cancer patients initiating aromatase inhibitor therapy or prostate cancer patients initiating androgen deprivation therapy; and (4) an age-matched non-cancer control cohort (n = 100) without cancer diagnosis, receiving no neurotoxic medication.

Baseline Assessment

All participants would undergo comprehensive baseline evaluation including: comprehensive neuropsychological testing battery (processing speed, executive function, attention, episodic memory, verbal fluency, and language); structural brain MRI with white matter volumetry; quantitative EEG with spectral power and coherence analysis; complete inflammatory panel (hs-CRP, IL-6, TNF-α, IL-1β); cardiac assessment including echocardiography and heart rate variability analysis (to enable discrimination of cardiac-origin

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cognitive changes); and patient-reported outcome measures for cognitive complaints (FACT-Cog), fatigue (FACIT-F), mood (PHQ-9, GAD-7), and quality of life (EORTC QLQ-C30).

Linguistic Data Collection

Participants would provide standardized narrative samples at the following frequencies: pre-treatment baseline (three sessions over two weeks to establish individual norms), active treatment phase (weekly, timed to capture both acute post-treatment perturbation and inter-cycle recovery—specifically, samples collected 48 hours post-treatment and 5–7 days post-treatment for each cycle), post-treatment recovery phase (biweekly for 6 months, then monthly for 18 months), and long-term survivorship (monthly for the duration of follow-up). Each session would include: a semi-structured autobiographical narrative (5-minute spoken), a picture description task, a written narrative sample (dream report or autobiographical narrative), and daily 60-second voice diary entries via smartphone for participants willing to provide high-frequency data.

All spoken samples would undergo both transcription-based linguistic analysis and acoustic-prosodic analysis. The prosodic analysis serves a specific differential diagnostic function in this study: it enables discrimination of cancer-related cognitive changes (predicted to lack prosodic-autonomic features) from cardiac-origin cognitive changes (predicted to show early prosodic disruption), particularly important in patients receiving potentially cardiotoxic cancer treatments (anthracyclines, trastuzumab) who may develop concurrent cardiac and cancer-related cognitive impairment.

Follow-Up Assessment

Full neuropsychological, neuroimaging, and biomarker assessment would be repeated at: end of active treatment, 6 months post-treatment, 12 months post-treatment, and 24 months post-

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treatment. Inflammatory markers would be assessed at each treatment cycle to enable correlation with linguistic trajectory data. Cardiac monitoring would be performed at baseline and end-of-treatment for patients receiving cardiotoxic regimens. Interim clinical events—including treatment dose modifications for toxicity, hospitalizations for neurological adverse events, imaging-detected brain metastases, and treatment discontinuation for cognitive complaints—would be documented as endpoints.

Primary Analyses

The proposed analyses address six sequential questions.

First, pre-treatment detection: at baseline, do cancer patients exhibit the predicted inflammatory-cognitive linguistic signature compared with age-matched controls? Does baseline linguistic impairment correlate with inflammatory biomarker levels? Does baseline linguistic status predict subsequent treatment-related cognitive decline?

Second, treatment-cycle sensitivity: do linguistic features show the predicted cyclic perturbation pattern synchronized with chemotherapy cycles? Is the magnitude of post-cycle linguistic perturbation dose-dependent and cumulative? Can the treatment-cycle-aware trajectory model distinguish expected from excessive neurotoxicity?

Third, immunotherapy surveillance: among immunotherapy patients, do those who develop neurological irAEs show detectable linguistic trajectory changes before clinical recognition? How far in advance of clinical diagnosis do these changes become statistically significant?

Fourth, endocrine therapy monitoring: do patients initiating aromatase inhibitors or androgen deprivation therapy show the predicted memory-predominant or spatial-executive

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linguistic trajectory changes? Do these changes correlate with hormone levels and can they discriminate endocrine-therapy-related cognitive changes from other etiologies?

Fifth, recovery trajectory characterization: following treatment completion, do linguistic trajectories show the predicted recovery patterns? Can the trajectory characteristics (rate, completeness, reversal) identify patients at risk for persistent CRCI?

Sixth, differential discrimination: can the proposed NOST-based cancer-related cognitive signature (treatment-cycle-synchronized, prosodic-autonomic-absent, domain-specific) be reliably distinguished from cerebrovascular, cardiac, psychiatric, and age-related cognitive decline in patients who may experience multiple concurrent causes of cognitive change?

Power Considerations

The chemotherapy cohort of 200 participants with weekly sampling during treatment (approximately 16–24 weeks depending on regimen) and biweekly-to-monthly sampling during follow-up provides approximately 8,000–10,000 linguistic data points during treatment and 4,000–6,000 during follow-up. Based on the prevalence of CRCI (30–75% during treatment, 20–35% persistent), the study would include approximately 60–150 patients with measurable cognitive impairment during treatment, providing adequate power for trajectory characterization. The immunotherapy cohort of 100 patients, with an expected neurological irAE rate of 1–5%, would yield 1–5 cases of neurological adverse events—sufficient for case-level demonstration of the detection framework but insufficient for formal sensitivity analysis, suggesting that multi-site collaboration would be needed for definitive validation of the immunotherapy surveillance application.

Clinical Implications

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Real-Time Neurotoxicity Monitoring

The most immediate and clinically actionable application is real-time monitoring of treatment-related neurotoxicity. Currently, oncologists have no objective, real-time method for assessing whether a patient's chemotherapy regimen is producing cognitive damage. Decisions about dose reduction, treatment delay, or regimen change for neurotoxicity are made reactively—after the patient or family reports cognitive complaints severe enough to prompt clinical attention. By this point, substantial cumulative neurotoxicity may have occurred. NOST-based monitoring would provide proactive neurotoxicity assessment: a continuous cognitive trajectory that alerts the treatment team when cumulative neurotoxicity exceeds expected bounds, potentially triggering dose modification before irreversible damage occurs.

Immunotherapy Safety Monitoring

Early detection of immunotherapy-related neurological adverse events is potentially life-saving. Autoimmune encephalitis, if detected early, is typically reversible with immunosuppressive treatment. If detected late, it can produce permanent cognitive disability or death. Current detection depends on clinical recognition of symptoms that are often initially subtle and easily attributed to other causes. Continuous linguistic monitoring would provide an independent detection channel that does not depend on patient self-report or clinician recognition—particularly valuable in the context of immunotherapy, where the spectrum of possible adverse events is broad and the clinical index of suspicion for neurological complications may be low.

Treatment Decision Support

NOST-based cognitive monitoring would provide objective data for several treatment decisions that currently rely on subjective assessment: whether to continue, modify, or

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discontinue a neurotoxic regimen based on the cognitive cost-benefit ratio; whether to select hippocampal-avoidance radiation techniques based on pre-treatment cognitive vulnerability; whether to continue endocrine therapy when a patient reports cognitive complaints (distinguishing objective cognitive decline from subjective complaints without measurable impairment); and whether to recommend cognitive rehabilitation, pharmacological neuroprotection, or behavioral strategies based on the specific cognitive domains affected.

Survivorship Care Planning

For the 18+ million cancer survivors in the United States, NOST-based cognitive monitoring addresses a critical gap in survivorship care. Current survivorship guidelines recommend cognitive screening but provide no scalable method for longitudinal cognitive tracking. Language-based monitoring through a smartphone application could provide continuous surveillance at minimal cost and burden, enabling: early detection of late cognitive effects years after treatment, identification of survivors whose cognitive trajectory suggests accelerated aging requiring intervention, objective documentation of cognitive impairment for survivors seeking disability accommodation or workplace support, and outcome monitoring for cognitive rehabilitation programs.

Relationship to the Vascular-Cardiac Surveillance Framework

The oncological surveillance framework integrates with but does not duplicate the cerebrovascular (Beverly, 2026e) and cardiac (Beverly, 2026f) surveillance frameworks. The three clinical applications share common processing infrastructure (REMIEL extraction, NOST mapping, NOSS scoring, longitudinal trajectory analysis) but apply distinct diagnostic templates, temporal models, and referral pathways.

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The critical integration point is the management of patients who may experience concurrent cancer-related and cardiovascular cognitive effects. Patients receiving anthracycline chemotherapy, for example, are at risk for both chemotherapy neurotoxicity (an oncological pathway) and cardiotoxicity-mediated cognitive impairment (a cardiac pathway). The Differential Signature Engine must parse the observed linguistic trajectory into its component etiological contributions—distinguishing chemotherapy-cycle-synchronized neurotoxicity from cardiac-output-dependent cognitive decline developing secondary to treatment-induced cardiomyopathy. The prosodic-autonomic discriminator (present in cardiac-origin decline, absent in pure cancer-related decline) provides the primary differentiating feature, but treatment-cycle temporal analysis and cardiac biomarker correlation provide additional discriminating evidence.

This multi-etiology parsing capability—determining whether an observed cognitive trajectory change reflects cancer-related neurotoxicity, treatment-induced cardiotoxicity, coincidental cerebrovascular disease, primary psychiatric comorbidity, or combinations thereof—represents the most technically challenging and clinically valuable aspect of the integrated NOUS Clinical Surveillance Platform.

Limitations and Caveats

Several significant limitations must be acknowledged.

First, as with the companion papers, this framework is entirely theoretical. No empirical data currently exist testing whether NOST-based linguistic monitoring detects cancer-related cognitive impairment. The proposed detection pathways are logically coherent and each link is supported by published evidence, but the integrated system has not been validated.

Second, cancer patients often experience multiple concurrent factors affecting cognition—fatigue, pain, medication effects (opioids, antiemetics, corticosteroids), sleep

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disruption, psychological distress—that may produce linguistic changes confounding the cancer-specific signal. The proposed framework must contend with substantially more noise than the cerebrovascular or cardiac applications, where the confounding cognitive influences are fewer.

Third, cancer treatment protocols are heterogeneous. Different chemotherapy regimens, radiation fields, immunotherapy agents, and endocrine therapies produce different patterns of cognitive impact. The proposed validation study covers major treatment categories but cannot comprehensively validate the framework for every possible treatment combination. Extensive post-validation work would be needed to calibrate expected cognitive trajectories for specific regimens.

Fourth, the immunotherapy surveillance application, while potentially the most clinically urgent, involves the rarest events (1–5% neurological irAE rate). Single-site studies will be underpowered for this application, and multi-site collaboration with large immunotherapy registries would be essential for definitive validation.

Fifth, the survivorship application depends on sustained patient engagement with longitudinal monitoring over years. Compliance with monthly language sample collection in the absence of active treatment or acute medical concern may prove challenging, and the minimum monitoring frequency required for clinically useful surveillance in the survivorship phase is an open empirical question.

Sixth, the author's commercial interest in NOST-based applications constitutes a potential bias. Independent validation by oncology research groups without commercial ties to the technology is essential. The proposed study would benefit from collaboration with NCI-designated comprehensive cancer centers and cooperative oncology groups with established neurocognitive research programs.

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Conclusion

This paper proposes that cancer-related cognitive impairment—across its full spectrum from pre-treatment inflammation-mediated baseline impairment through treatment-related neurotoxicity to long-term survivorship effects—produces neural oscillatory changes detectable through NOST-based linguistic analysis. Six pathways connecting cancer and its treatment to neural oscillatory disruption are identified, each producing predicted linguistic signatures with distinct temporal dynamics, domain-specific profiles, and treatment implications.

The clinical need is urgent and the detection gap is real. Cancer-related cognitive impairment affects millions of patients, degrades quality of life and functional independence, and has no current objective, scalable monitoring solution. If NOST-based linguistic surveillance proves valid in the oncological context, it would provide the first continuous cognitive monitoring tool in cancer care—enabling real-time neurotoxicity detection, treatment optimization, immunotherapy safety monitoring, and long-term survivorship cognitive tracking.

The framework is offered for scientific scrutiny. The proposed validation study provides a structured research program with testable predictions, specified outcome measures, and defined success criteria. The integration of oncological surveillance with the vascular-cardiac surveillance framework described in companion papers creates a comprehensive NOUS Clinical Surveillance Platform capable of detecting the cognitive consequences of multiple disease processes through a single, non-invasive linguistic monitoring modality.

The scientific community's engagement—particularly from oncologists, neuro-oncologists, and cancer survivorship researchers—is essential and welcomed.

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